ACTIVITY-1

Agile Methodology vs Waterfall Methodology in Project Management:

Agile Project Management	**Waterfall Project Management**
Client input is required throughout the product development.	Client input is required only after completing each phase.
Changes can be made at any stage.	Changes can be made at any stage.
Coordination among project teams is required to ensure correctness.	Coordination is not needed as one team starts the work after the finish of another team.
It is really useful in large and complex projects.	It is mainly used for small project development.
The testing part can be started before the development of the entire product.	Testing can only be performed when the complete product is ready.
A Small team is sufficient for Agile project management.	It requires a large team.
The cost of development is less.	The cost of development is high.
It completes the project in comparatively less time.	It takes more time compared to Agile.
The Agile Method is known for its flexibility.	The waterfall Method is a structured software development methodology, so it is quite rigid.

ACTIVITY 2

What Is the Agile Manifesto?

The Agile Manifesto is a foundational document created in 2001 by 17 software experts at Snowbird, USA. They wanted a better way to build software because traditional methods (like Waterfall) were slow and rigid.

The manifesto contains:

A. Four Core Values

1.Individuals and interactions over processes and tools

• People and communication produce better results than strict rules.

• Example: A quick discussion with a developer is better than writing long emails or documents.

2.Working software over comprehensive documentation

• Deliver a working product quickly.

• Documentation is necessary but should not delay delivery.

3. Customer collaboration over contract negotiation

• Customers should give regular feedback.

• Teams and customers work closely to improve the product continuously.

4.Responding to change over following a rigid plan

• Change is natural in software.

• Agile teams accept change even late in development.

B. Twelve Agile Principles

1. Highest priority is customer satisfaction.
2. Welcome changing requirements anytime.
3. Deliver working software frequently (every 1–4 weeks).
4. Business and developers work together daily.
5. Build projects around motivated individuals.
6. Face-to-face communication is most effective.
7. Working software is the main measure of progress.

ACTIVITY 3

Tools used in scrum process

Scrum is not a tool; it is a framework used in Agile project management, particularly in software development. It provides a structured approach to managing complex projects by breaking them into smaller, manageable iterations called sprints. Scrum focuses on collaboration, adaptability, and delivering value incrementally.

Key Characteristics of Scrum

Scrum defines specific roles (e.g., Scrum Master, Product Owner, Development Team), events (e.g., Sprint Planning, Daily Stand-ups, Sprint Retrospectives), and artifacts (e.g., Product Backlog, Sprint Backlog) to ensure efficient project execution.

While Scrum itself is not a tool, it often relies on tools to facilitate its processes. These tools help manage tasks, track progress, and improve team collaboration.

Tools Commonly Used in Scrum

- Jira: For managing backlogs, tracking sprints, and visualizing workflows.
- Trello: For task management and workflow visualization.
- Confluence: For documentation and knowledge sharing.
- Slack: For real-time communication.

Skills Required for Scrum

Implementing Scrum effectively requires skills such as collaboration, facilitation, prioritization, and adaptability. These skills are essential for roles like Scrum Master and Product Owner to ensure smooth execution of the framework.

In conclusion, Scrum is a framework, not a tool or skill. However, it requires specific skills to implement and can be supported by various tools to streamline its processes.

ACTIVITY 4

1. BLACK BOX TESTING

Black Box Testing is a software testing technique where the tester does not know the internal logic, code, or structure of the application.

It is also called Functional Testing because it focuses only on what the software *does*, not how it works internally.

This method checks the external behaviour of the application from the user's point of view.

Example (Amazon App):

Search "iPhone 14" → Verify whether the correct product results appear.

You do not know how the search algorithm works internally.

Common Techniques

1. **Equivalence Partitioning** – Divides input data into valid/invalid classes to reduce test cases.
2. **Boundary Value Analysis** – Focuses on values at the edges of input ranges.
3. **Decision Table Testing** – Uses cause-effect combinations to derive scenarios.
4. **State Transition Testing** – Validates behavior changes between states.
5. **Error Guessing** – Leverages tester experience to find likely defect areas.

Example Testing a login form:

- Valid input: Correct username/password → Successful login.
- Invalid input: Wrong password → Error message.
- Empty fields: Prompt for required input.

No backend code is reviewed; only the behavior is validated.

Advantages

- No programming knowledge required.
- Independent and unbiased testing.

ACTIVITY 5

1. Equivalence Partitioning (EP)

Equivalence Partitioning groups inputs into valid and invalid categories so that a single test from each group represents the entire set.

ATM Withdrawal – Input Partitions

Input Condition	Valid Group	Invalid Group
Card Status	Active / usable card	Blocked or expired card
PIN Entry	Correct PIN	Incorrect or mistyped PIN
Withdrawal Amount	Amount within allowed limits	Below minimum / above maximum
Account Funds	Enough balance	Insufficient balance

✓ Five EP Test Cases

TC No	Scenario	Input	Expected Output
EP01	Valid card, correct PIN, acceptable amount	Card Valid, PIN OK, Amount ₹500	Cash withdrawal successful
EP02	Expired or blocked card	Card Expired	ATM declines the card
EP03	Wrong PIN entry	Enter Incorrect PIN	"Incorrect PIN" prompt
EP04	Amount lower than allowed minimum	₹50 withdrawal attempt	Error: Minimum allowed is ₹100
EP05	High withdrawal with low account balance	Withdraw ₹10,000; Balance ₹2,000	"Insufficient balance" message

2. Boundary Value Analysis (BVA)

BVA focuses on testing edge values where errors are most likely to occur.

ATM Rules